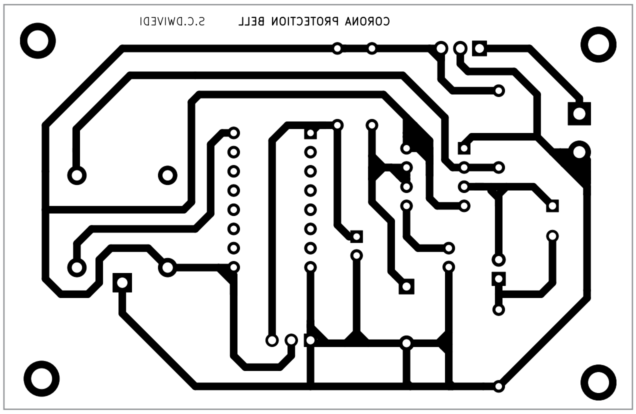

Fig. 5: Actual-size PCB of the circuit

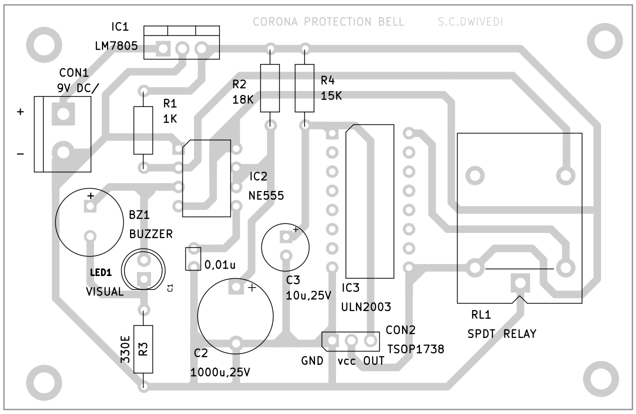

Fig. 6: Component layout of the PCB

PARTS LIST	
Semiconductors:	
IC1	- LM7805, 5V voltage regulator
IC2	- NE555 timer
IC3	- ULN2003 relay driver
LED1	- 5mm LED
Resistors (all 1/4-watt, ±5% carbon):	
R1	- 1-kilo-ohm
R2	- 18-kilo-ohm
R3	- 330-ohm
R4	- 15-kilo-ohm
Capacitors:	
C1	- 0.01µF ceramic disk
C2	- 1000µF, 25V electrolytic
C3	- 10µF, 25V electrolytic
Miscellaneous:	
CON1	- 2-pin connector
BZ1	- Buzzer
RL1	- 5V SPDT relay
	- TSOP1738 sensor
	- 9V battery/9V DC

receive the signals from their remote controls. TSOP1738 is a commonly used infrared receiver; it has a PIN diode, preamplifier, and internal filter for the PCM frequency. The carrier frequency of TSOP1738 is 38kHz. An image of TSOP1738 is shown in Fig. 3.

Relay driver ULN2003 (IC3) is used here to drive 5V relay RL1. Pin diagram of ULN2003 is shown in Fig. 4.

Working of the circuit is simple. The remote may be fixed or kept near the wash basin. Whenever anyone goes to wash hands, the remote may be used to switch on the LED and the buzzer for 20 seconds, so the person can rub hands with soap till the LED glows and the buzzer sounds. After that the person may wash the hands with water.

An actual-size, single-side PCB layout for the circuit is shown in Fig. 5 and its component layout in Fig. 6. After assembling the circuit on PCB, enclose it in a suitable box.

Fix TSOP1738, LED1, and CON1 in front of box. Fit the 9V battery inside the box and the buzzer either inside or on back side of the box. Make a hole in front of TSOP178 in such a way that when you press remote its signal should fall on it. EFY

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